EEG Patterns in Alcoholic and Control Participants

Spatial participant classification and visual stimulus-condition prediction

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INTRODUCTION & QUESTIONS

EEG records scalp voltage over time. We test whether these recordings contain patterns related to participant group and the visual recognition task.

Who? Can EEG distinguish alcoholic from control participants?

How? Can EEG identify the stimulus condition, and does predictive performance differ by group?

ORIGINAL DATASET

20 participants: 10 alcoholic, 10 control.
600 trials: 30 per participant; 200 per condition.
64 recorded channels × 256 samples per trial; 256 Hz per UCI documentation.

S1: single stimulus
S2 match: matching second stimulus
S2 non-match: different second stimulus

METHODS & VALIDATION

1. Validate sample and trial identities; keep all 600 original trials.
2. Apply a 1–40 Hz filter; extract spatial or temporal/spectral features.
3. Split by participant, not trial. Use 5 outer folds and 3 inner folds.
4. Fit imputation, scaling, selection/PCA and classifiers on training participants only.
5. Repeat with seeds 183, 184 and 185. These reuse the same 20 people.

CONCLUSIONS & LIMITATIONS

Retain the spatial participant model. Tested frequency/geometry and 18 spatial adjustments did not exceed its 85% mean.

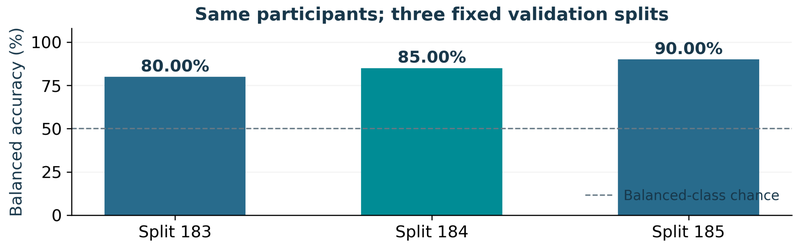
Only 20 people; repeated development on the same cohort. No independent external test.

Stimulus onset is unverified: recording-relative traces are not confirmed ERPs. Classifier probabilities are uncalibrated.

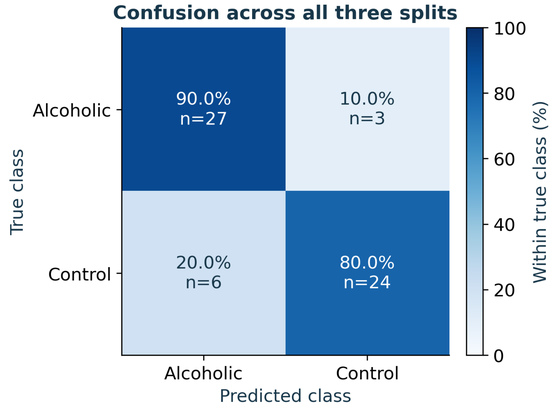
MODEL 1 | PARTICIPANT CLASSIFIER

85% mean balanced accuracy

Alcoholic vs control | 20 held-out people per split
Spatial selection: log covariance/correlation, condition-wise aggregation, PCA (6), regularized logistic regression.



Above 50% balanced-class chance. Split accuracies: 80%, 85%, 90%. Macro-F1 mean: 0.850.



60 held-out decisions from 20 participants across 3 splits; repeated decisions are not independent people. Pooled recall: alcoholic 90%; control 80%.

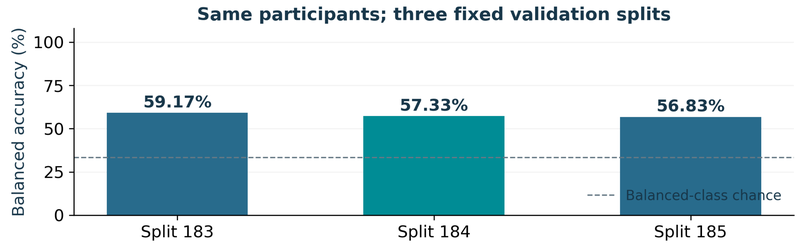
Final saved spatial configuration
61 scalp channels → OAS log covariance → mean per condition → training-fold scaling → PCA 6 → logistic regression (C=0.1). X, Y and nd are excluded from the spatial branch.

Interpretation: EEG features distinguish dataset labels in this subset. This is not a validated clinical diagnostic test or proof that alcoholism caused the observed patterns.

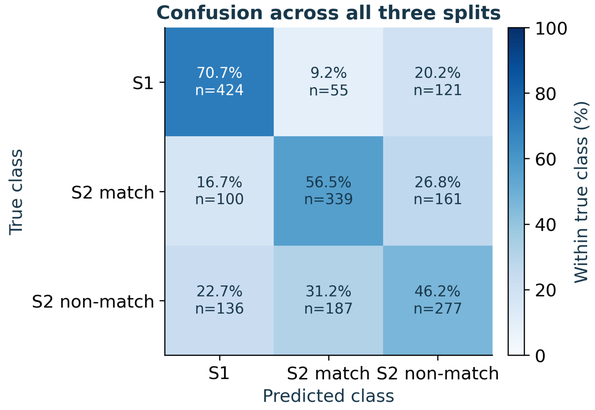
MODEL 2 | STIMULUS-CONDITION MODEL

57.78% mean balanced accuracy

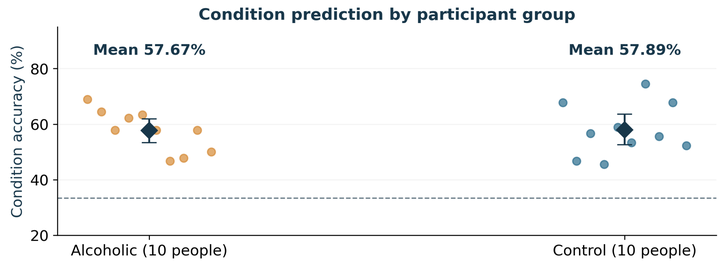
S1 vs S2 match vs S2 non-match | 600 held-out trials per split
Temporal/spectral candidate selection; saved refit uses 8-bin features, selection of 256 features and logistic regression.



Above 33.33% balanced-class chance. Split accuracies: 59.17%, 57.33%, 56.83%. Macro-F1 mean: 0.575.



1,800 held-out decisions from 600 trials across 3 splits. Pooled recall: S1 70.7%; match 56.5%; non-match 46.2%.



Alcoholic: 57.67%; control: 57.89%. Dots represent 10 participants per group; diamonds show means with exploratory 95% participant bootstrap intervals.

HOW question: condition decoding is feasible, but these nearly equal group accuracies do not establish different neural responses.

Sources: UCI EEG Database (archive.ics.uci.edu/dataset/121/eeg+database); MDST Week 1 slides; original-data nested prediction logs.

Acknowledgments: MDST project leads Manishika Balamurugan, Panna Akter and Swetha Chimalakonda. Poster draft | 11 October 2026 | 48 × 36 in.